

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Develop a Styled Webpage Using HTML5 Canvas and CSS3

2. AIM

To develop a styled webpage using HTML5 Canvas and CSS3 that demonstrates basic graphics, layouts, styling properties, and interactive hover effects.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the fundamentals of HTML5 Canvas and CSS3.
 - To create basic graphics such as rectangles, circles, lines, and text using Canvas.
 - To understand and apply CSS selectors and syntax.
 - To style webpage elements using colors, fonts, borders, margins, and padding.
 - To apply CSS3 features such as gradients, shadows, rounded corners, and hover effects.
 - To design an attractive webpage using a header, navigation bar, content cards, and footer.
-

4. THEORY

HTML5 is the latest version of HTML that provides improved structure, multimedia support, graphics, and modern web features.

CSS3 is a stylesheet language used to control the appearance and layout of HTML elements. It provides features such as colors, fonts, borders, gradients, shadows, rounded corners, and hover effects.

HTML5 Canvas is an element used to draw graphics dynamically using JavaScript. It can be used to draw shapes, lines, and text.

In this application:

- `<canvas>` is used to draw a rectangle, circle, line, and text.
- CSS selectors are used to target and style HTML elements.
- Colors and fonts are used to improve the visual appearance of the webpage.
- Margin and padding are used to control spacing between and inside elements.
- Borders and border-radius are used to define and style element boundaries.
- Gradients are used to create smooth color transitions.
- Box shadow and text shadow are used to create depth and emphasis.
- Hover effects are used to provide visual feedback when the user interacts with buttons and navigation links.

The basic structure of the webpage is:

HTML Structure → Canvas Graphics → CSS Styling → Interactive Hover Effects → Styled Webpage

5. ALGORITHM / PROCEDURE

1. Create a new HTML file and define the HTML5 document structure.
2. Add a header, navigation bar, three content cards, and a footer.
3. Add a `<canvas>` element to the webpage.
4. Use JavaScript to draw a rectangle, circle, line, and name on the canvas.
5. Create CSS selectors to style the HTML elements.
6. Apply background colors, gradient backgrounds, fonts, borders, margins, and padding.
7. Apply rounded corners using border-radius.
8. Add box-shadow and text-shadow effects to the required elements.
9. Add hover effects to navigation links and buttons.
10. Save the HTML and CSS code and open the webpage in a browser.
11. Verify that the Canvas graphics and all CSS styles are displayed correctly.
12. Test the hover effects by moving the cursor over the navigation links and buttons.

13. Compare the actual output with the expected design and make necessary corrections.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Creative Webpage</title>

  <link rel="stylesheet" href="style.css">
</head>

<body>

  <!-- Header -->
  <div class="header">
    <h1>Creative Webpage</h1>
    <p>Exploring HTML5 Canvas and CSS3 Design</p>
  </div>

  <!-- Navigation -->
  <div class="navbar">
    <a href="#">Home</a>
    <a href="#">Gallery</a>
    <a href="#">Projects</a>
    <a href="#">Contact</a>
  </div>

  <!-- Hero Section -->
  <div class="hero">
    <h2>Welcome to My Webpage</h2>
    <p>
```

```
        This webpage demonstrates the use of HTML5 Canvas,  
        responsive layouts, and CSS styling techniques.  
    </p>  
</div>  
  
<!-- Canvas Section -->  
<div class="canvas-section">  
    <h2>HTML5 Canvas Demonstration</h2>  
  
    <canvas id="myCanvas" width="700" height="250">  
        Your browser does not support HTML5 Canvas.  
    </canvas>  
</div>  
  
<!-- Cards -->  
<div class="cards-container">  
  
    <div class="card">  
        <h3>Canvas Graphics</h3>  
        <p>  
            HTML5 Canvas allows drawing shapes, lines, text,  
            and animations using JavaScript.  
        </p>  
  
        <button>Explore</button>  
    </div>  
  
    <div class="card">  
        <h3>Modern CSS</h3>  
        <p>  
            CSS3 provides responsive layouts, transitions,  
            gradients, shadows, and attractive designs.  
        </p>  
  
        <button>Learn More</button>  
    </div>  
  
    <div class="card">  
        <h3>Web Development</h3>  
        <p>
```

```
        Combining HTML, CSS, and JavaScript helps create
        interactive and visually appealing websites.
    </p>

    <button>Discover</button>
</div>

</div>

<!-- Footer -->
<div class="footer">
    <p>© 2026 Creative Webpage | Designed by Sravan Kumar</p>
</div>

<!-- Canvas Script -->
<script>

    const canvas = document.getElementById("myCanvas");
    const ctx = canvas.getContext("2d");

    // Background
    ctx.fillStyle = "#f4f4f4";
    ctx.fillRect(0, 0, canvas.width, canvas.height);

    // Rectangle
    ctx.fillStyle = "#4CAF50";
    ctx.fillRect(40, 40, 140, 80);

    // Circle
    ctx.beginPath();
    ctx.arc(280, 80, 40, 0, Math.PI * 2);
    ctx.fillStyle = "#2196F3";
    ctx.fill();

    // Triangle
    ctx.beginPath();
    ctx.moveTo(420, 120);
    ctx.lineTo(500, 40);
    ctx.lineTo(580, 120);
    ctx.closePath();
```

```

        ctx.fillStyle = "#FF9800";
        ctx.fill();

        // Line
        ctx.beginPath();
        ctx.moveTo(40, 180);
        ctx.lineTo(650, 180);
        ctx.strokeStyle = "#555";
        ctx.lineWidth = 2;
        ctx.stroke();

        // Name
        ctx.font = "26px Arial";
        ctx.fillStyle = "#E91E63";
        ctx.fillText("Designed by Sravan Kumar", 180, 225);

</script>

</body>

</html>

```

style.css

```

/* =====
   Universal Selector
===== */
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
}
/* =====
   Body
===== */
body{
    font-family:Arial, Helvetica, sans-serif;
    background:#111315;
    color:#e8e8e8;
}

```

```
        line-height:1.6;
    }
    /* =====
        Header
    ===== */
    .header{
        text-align:center;
        padding:35px;
        background:#181a1d;
        border-bottom:3px solid #d90429;
    }

    .header h1{
        font-size:42px;
        color:#ffffff;
        letter-spacing:2px;
    }
    .header p{
        margin-top:10px;
        color:#bdbdbd;
        font-size:17px;
    }
    /* =====
        Navigation
    ===== */
    .navbar{
        display:flex;
        justify-content:center;
        background:#0d0d0d;
        border-bottom:1px solid #2b2b2b;
    }
    .navbar a{
        color:#f5f5f5;
        text-decoration:none;
        padding:18px 25px;
        transition:0.3s;
    }
    .navbar a:hover{
        color:#d90429;
        border-bottom:2px solid #d90429;
```

```

}
/* =====
    Hero
===== */
.hero{
    width:90%;
    margin:30px auto;
    background:#1a1d21;
    border-left:5px solid #d90429;
    padding:35px;
    border-radius:10px;
}
.hero h2{
    color:#ffffff;
    margin-bottom:15px;
}
.hero p{
    color:#c9c9c9;
}
/* =====
    Canvas Section
===== */
.canvas-section{
    width:90%;
    margin:30px auto;
    background:#1a1d21;
    padding:25px;
    border-radius:10px;
    border:1px solid #2b2b2b;
    text-align:center;
}
.canvas-section h2{
    color:#ffffff;
    margin-bottom:20px;
}
canvas{
    background:white;
    border-radius:8px;
    border:2px solid #d90429;
}

```

```

/* =====
    Cards
===== */
.cards-container{
    width:90%;
    margin:auto;
    display:flex;
    gap:20px;
    flex-wrap:wrap;
}
.card{
    flex:1;
    min-width:250px;
    background:#1a1d21;
    border:1px solid #2b2b2b;
    border-radius:12px;
    padding:25px;
    transition:.3s;
}
.card:hover{
    transform:translateY(-6px);
    border-color:#d90429;
}
.card h3{
    color:#ffffff;
    margin-bottom:15px;
}
.card p{
    color:#c9c9c9;
    margin-bottom:20px;
}
/* =====
    Buttons
===== */
button{
    padding:10px 20px;
    border:none;
    border-radius:6px;
    background:#d90429;
    color:white;
}

```

```
        cursor:pointer;
        transition:.3s;
    }
    button:hover{
        background:#b00020;
        transform:translateY(-2px);
    }
    /* =====
        Footer
    ===== */
    .footer{
        margin-top:30px;
        padding:20px;
        text-align:center;
        background:#0d0d0d;
        color:#9a9a9a;
        border-top:2px solid #d90429;
    }
    /* =====
        Responsive
    ===== */
    @media (max-width:768px) {
        .navbar{
            flex-direction:column;
            align-items:center;
        }
        .cards-container{
            flex-direction:column;
        }
    }
}
```

7. CODE EXPLANATION

Code	Explanation
<code><!DOCTYPE html></code>	Declares the document as an HTML5 document.
<code><html lang="en"></code>	Defines the root element and specifies English as the document language.
<code><head></code>	Contains metadata, the page title, and the CSS stylesheet reference.
<code><meta charset="UTF-8"></code>	Specifies UTF-8 character encoding.
<code><meta name="viewport"></code>	Helps the webpage adapt to different screen sizes.
<code><title></code>	Sets the title displayed in the browser tab.
<code><link rel="stylesheet" href="style.css"></code>	Connects the external CSS file to the HTML document.
<code><div class="header"></code>	Creates the header section of the webpage.
<code><div class="navbar"></code>	Creates the navigation bar containing navigation links.
<code></code>	Creates a hyperlink.
<code><div class="hero"></code>	Creates the main introductory section of the webpage.
<code><div class="canvas-section"></code>	Creates the section containing the HTML5 Canvas.
<code><canvas id="myCanvas" width="700" height="250"></code>	Creates a 700 × 250 pixel drawing area for Canvas graphics.
<code><div class="cards-container"></code>	Contains the three content cards.
<code><div class="card"></code>	Defines an individual content card.

<code><button></code>	Creates an interactive button for each card.
<code><div class="footer"></code>	Creates the footer section of the webpage.
<code><script></code>	Contains the JavaScript code used to draw graphics on the Canvas.
<code>document.getElementById("myCanvas")</code>	Retrieves the Canvas element using its ID.
<code>canvas.getContext("2d")</code>	Obtains the 2D drawing context used to draw on the Canvas.
<code>ctx.fillStyle</code>	Sets the color used for filling shapes or text.
<code>ctx.fillRect()</code>	Draws a filled rectangle on the Canvas.
<code>ctx.beginPath()</code>	Starts a new drawing path.
<code>ctx.arc()</code>	Creates a circular arc, which is used to draw the circle.
<code>ctx.fill()</code>	Fills the current path with the selected color.
<code>ctx.moveTo()</code>	Moves the drawing cursor to a specified position without drawing.
<code>ctx.lineTo()</code>	Creates a line from the current position to the specified position.
<code>ctx.closePath()</code>	Closes the current drawing path.
<code>ctx.stroke()</code>	Draws the outline of the current path.
<code>ctx.strokeStyle</code>	Sets the color of the line.
<code>ctx.lineWidth</code>	Sets the thickness of the line.
<code>ctx.font</code>	Specifies the font style and size used for Canvas text.

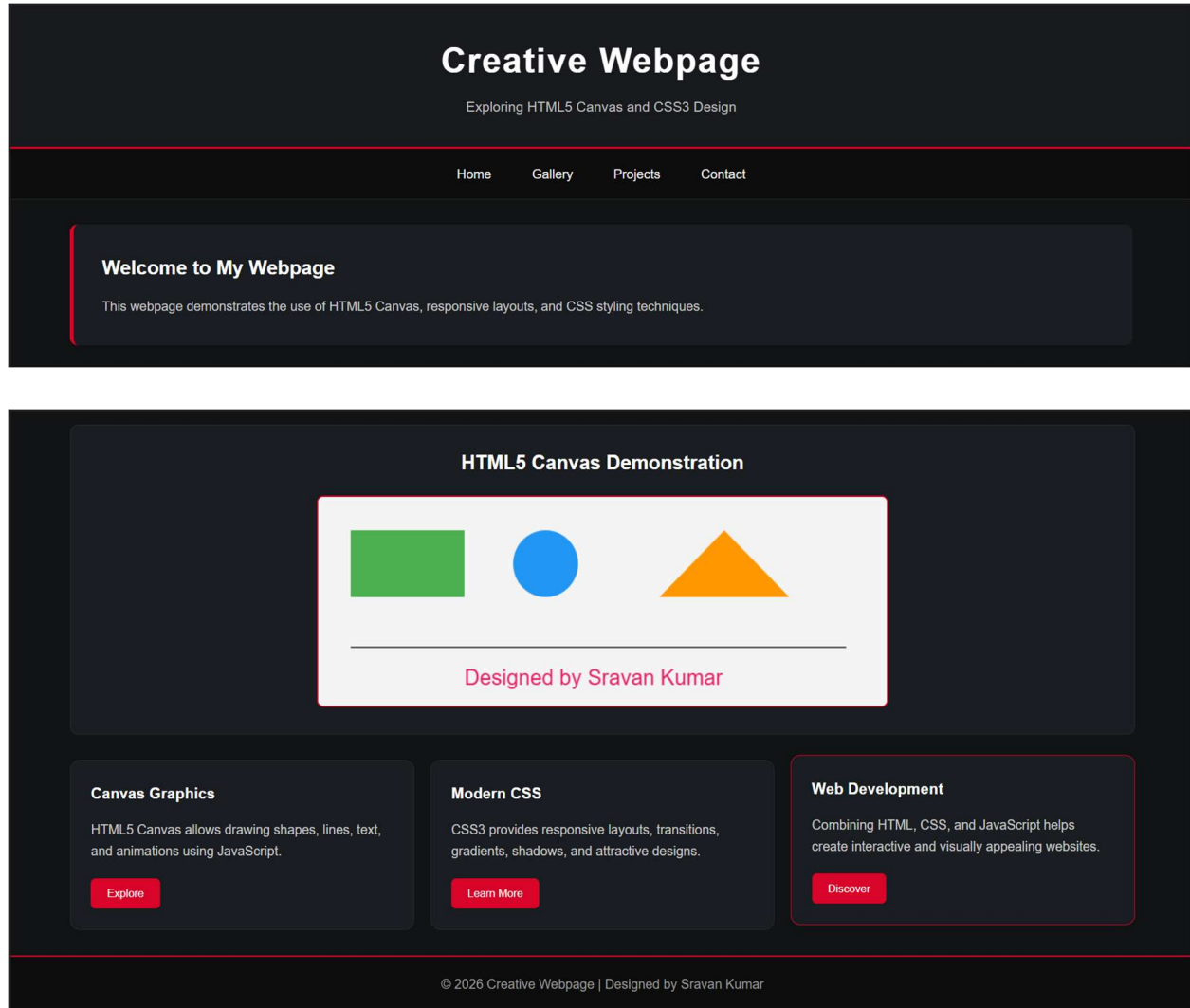
<code>ctx.fillText()</code>	Displays text at a specified position on the Canvas.
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8. TEST CASES AND EXECUTION DATA

Test Case	Expected Result	Actual Result	Status	Test Case	Expected Result	Actual Result
Open the HTML file in a browser	Webpage loads with header, navigation, cards, and footer	Webpage loaded correctly	Pass	Open the HTML file in a browser	Webpage loads with header, navigation, cards, and footer	Webpage loaded correctly
Check Canvas graphics	Rectangle, circle, triangle, line, and name are displayed	All graphics displayed correctly	Pass	Check Canvas graphics	Rectangle, circle, triangle, line, and name are displayed	All graphics displayed correctly
Hover over navigation links	Link color and border change smoothly	Hover effect works correctly	Pass	Hover over navigation links	Link color and border change smoothly	Hover effect works correctly
Hover over content cards	Card moves upward and border changes color	Card hover effect works correctly	Pass	Hover over content cards	Card moves upward and border changes color	Card hover effect works correctly

9. OUTPUT

Output 1: Static web page with css and canvas applied



10. RESULT

Thus, a styled webpage was successfully developed using HTML5 Canvas and CSS3. Basic graphics, responsive layout, CSS3 styling properties, and interactive hover effects were implemented successfully, and the webpage was verified using multiple test cases.